HDL is also called **'good cholesterol'**, because it prevents atherosclerosis. High HDL levels are associated with lowered risk of CVD. HDL level below 40 mg/dl of blood is considered to be a major risk factor for developing heart disease whereas HDL levels of 60 mg/dl or more greatly lowers the risk for CVD (NCEP, 2002).

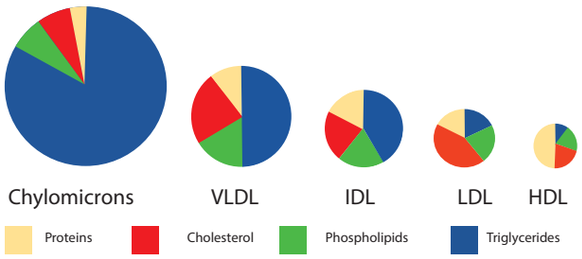


Fig 11.5: Types of lipoproteins

A plaque is a fatty deposit of cholesterol within the arterial wall that accumulates over a period of time. It may grow in size eventually blocking the artery and cutting off the blood supply to the tissue.

11.11 Risk Factors for CVD

Risk factors for CVD may be categorised as Non-modifiable risk factors and Modifiable risk factors :

I) Non-modifiable risk factors for CVD

Non-modifiable risk factors are those that cannot be changed. For CVD, **the most important non-modifiable risk factors are as follows:**

1. **Age:** Advancing age increases the risk of developing heart disease. It occurs mostly in men over 55 years of age and women over 65 years of age. Nowadays, CVD is seen in people in their 30s also.
2. **Gender:** Men are at a higher risk of developing CVD than women until women reach menopause. Post-menopausal women are at a similar risk as men.
3. **Ethnicity:** Risk of CVD changes around the world among ethnic groups. South Asians and African-Americans are at higher risk. Both genetics and lifestyle are likely to play a part.
4. **Family history:** Having a parent or sibling with a history of CVD increases the risk of developing CVD.

II) Modifiable risk factors for CVD

Modifiable risk factors are those which can be changed. By changing attitudes and practices, the risk of developing CVD can be reduced.

The major modifiable risk factors for CVD are:

1. **Hypertension:** Hypertension is the most common risk factor for CVD. If ignored, it can lead to stroke, coronary heart disease, angina, heart attack, and congestive heart failure. A diet high in salt and fat and low in fibre can contribute to the development of hypertension. Smoking, excessive alcohol consumption and a lack of physical activity can result in obesity and hypertension.
2. **Diabetes:** Insulin is required to carry glucose into the cells in order to be utilized for the release of energy. In diabetes, either the body does not produce sufficient insulin or the cells are unable to use it. This results in high circulating levels of blood glucose which may damage small blood vessels, mainly of the eyes, kidneys, and heart. Uncontrolled diabetes coupled with atherosclerosis further raises the risk of CVD.
3. **Elevated blood cholesterol levels:** High levels of LDL cholesterol contribute to plaque formation within the arteries, increasing the risk of heart attack. HDL cholesterol reduces the risk of CVD as it removes cholesterol from the arteries and takes it to the liver. High levels of HDL and low levels of LDL cholesterol lower risk of

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